

Proofs and Problem Solving

Week 9: Counting

Notes based on Martin Liebeck's

A Concise Introduction to Pure Mathematics

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15. Counting

Though combinatorics has been successfully applied to many branches of mathematics, these cannot be compared either in importance or in depth to the applications of analysis in number theory or algebra to topology, but I hope that time and the ingenuity of the younger generation will change this.

Paul Erdős, 1970.

This week we will introduce *combinatorics*, which is the study of counting. As you can see from the above quotation, Paul Erdős¹ felt that combinatorics played second fiddle to the fields of analysis and algebra. However, with the advent of computer science, combinatorics became essential for building efficient algorithms, and thus has a much larger importance for society than Erdős could have anticipated back in 1970.

If you enjoy the material this week, you may enjoy the course *Combinatorics and Graph Theory*.

15.1 The Multiplication Principle

Suppose I toss a coin twice: the total number outcomes is 4, since there are 2 outcomes for what the first coin can be (heads or tails), and then 2 outcomes for what the second coin

¹Paul Erdős was one of the greatest combinatorists in history, and the most prolific mathematician, publishing at least 1,525 mathematical papers, more than any other mathematician (though Euler wrote more total pages). His life is incredibly interesting as well, as he spent a large part of it technically homeless, with no permanent address, just travelling and staying with other mathematicians. For a short summary, you can read here (<http://www-history.mcs.st-and.ac.uk/Biographies/Erdos.html>), otherwise his biography *The Man Who Loved Only Numbers* is a great read.

can be. Hence the overall outcomes are (T,T), (T,H), (H,T), and (H,H), of which there are 4.

We can generalise this to events that involve more than just two possible outcomes: if I roll a 6-sided die and then an 8-sided die, the total number of outcomes is $6 \cdot 8 = 48$. I can figure this out because, for each of the outcomes for the first die (either 1, 2, ..., or 6), there are 8 possible outcomes for the second die, and so the total of all possible outcomes is 48. We can generalise this idea using the *multiplication principle*.

Theorem 15.1 — Multiplication Principle. Let P be a process consisting of n stages such that at each stage i , there are a_i choices we can make, and no two distinct choices can result in the same outcome. Then after n stages, the total number of outcomes is $a_1 \cdot a_2 \cdots a_n$.

Proof. We prove this by induction on n . For $n \in \mathbb{N}$, let $P(n)$ be the claim that, if P is a process consisting of n stages such that at each stage i , there are a_i choices we can make, then there are $a_1 \cdots a_n$ outcomes.

Base case: If $n = 1$, then P is a process of one stage where there are just a_1 choices, and so there are only a_1 many outcomes. This proves the base case.

Inductive step: Suppose $P(n)$ is true for some $n \in \mathbb{N}$. Suppose now P is a process with $n + 1$ stages and the i th stage has a_i choices. Note that if we stop the process before the last stage, we will have performed a process with n stages and for each $i = 1, \dots, n$, there are a_i choices. Then the induction hypothesis implies that there are $a_1 \cdots a_n$ outcomes. However, for each outcome of the first n stages, there are now a_{n+1} choices we can make. So the total number of outcomes after the $(n + 1)$ th stage is the total number of outcomes after n steps multiplied by the total number of outcomes that can happen from each of those outcomes, that is $(a_1 \cdots a_n) \cdot a_{n+1}$. This proves $P(n + 1)$. Thus, since $P(1)$ holds and $P(n) \implies P(n + 1)$ for each $n \in \mathbb{N}$, we conclude that $P(n)$ is true for all $n \in \mathbb{N}$ by induction. ■

This seems quite abstract, but let's look at a simple example.

Proposition 15.2 Let S have n elements. Then the number of subsets of S is 2^n .

Proof. Let S be a set of n elements and list the elements as $S = \{s_1, \dots, s_n\}$.

Let us define a process for picking a subset A of S : at each stage $i = 1, 2, \dots, n$, we decide whether to add s_i to our set or exclude it. At the end of the process, we will have picked a subset of A , and every subset of A is the outcome of such a process. By the Multiplication Principle, since there are 2 choices we can make at each stage, there are $2 \cdot 2 \cdots 2 = 2^n$ many outcomes for what A we could have chosen. Thus, there are 2^n sets. ■

If S is a finite set consisting of exactly n distinct members, we call n the *cardinality* or *size* of S and we denote this by $|S| = n$ (or in some texts by $\#(S) = n$).

Definition 15.1 Let $S = \{a_1, \dots, a_n\}$ be a set of n distinct objects. An *ordering* (or *arrangement*) of S is a sequence $(a_1, \dots, a_n)$ in which each element of S appears exactly once.

For example $(1, 3, 2)$ and $(3, 2, 1)$ are orderings of $\{1, 2, 3\}$. We can use the multiplication principle to count the number of orderings of S :

Example 15.1 How many ways are to order the set $\{1, 2, 3\}$?

What we do is develop a process for constructing all of the orderings. Suppose we would like to choose an ordering (a_1, a_2, a_3) where the a_i are distinct integers in $\{1, 2, 3\}$. We divide the process into 3 stages, where in the i th stage we pick the number a_i for our ordering.

Stage 1: There are 3 choices for a_1 : either 1, 2, or 3.

Stage 2: Having chosen a_1 , there are now only 2 numbers remaining that we could pick to be a_2 . So there are 2 choices for a_2 .

Stage 3: Having chosen a_1 and a_2 there is now only one choice for a_3 .

Thus, by the multiplication principle, there are $3 \cdot 2 \cdot 1 = 6$ outcomes, that is, 6 orderings.

In fact, more generally, we have the following:

Theorem 15.3 — Ordering Theorem. Given a set S of n distinct objects, there are $n!$ ways of ordering them.

This is a special case of the following proposition, (part (b) with $k = n$).

Proposition 15.4 Let S be a set with n elements and let $0 \leq k \leq n$.

- (a) The number of ways of picking an ordered selection of k elements from S (i.e. an ordered list of elements $(a_1, \dots, a_k)$, possibly with repetitions, with $a_i \in S$) is n^k .
- (b) The number of ways of picking an ordered selection of k *distinct* elements from S is $n(n-1) \cdots (n-k+1) = \frac{n!}{(n-k)!}$.

Proof. (a) We count these possibilities using a k -step process: there are n options for a_1 , n options for a_2 , and so on. The Multiplication Principle tells us that there are $n \cdot n \cdots n = n^k$ outcomes.

- (b) Again, we use the multiplication principle as in (a), but while there were n options for a_1 , now there are $n-1$ options for a_2 because, for the a_i to be distinct, we cannot repeat, so $a_2 \neq a_1$ and so there are only $n-1$ options for a_2 . Similarly, for a_3 there are $n-2$ options for what it can be (since it can't equal a_1 or a_2). Continuing in this way, after having picked $a_1, \dots, a_{i-1}$, there will be $n-i+1$ remaining options for a_i . Thus, the total number of sequences $a_1, \dots, a_k$ of k *distinct* elements of S is $n(n-1) \cdots (n-k+1)$. ■

15.2 Correspondence and overcounting

Sometimes we may try to count the members of some set, but we accidentally count the same elements more than once. We can avoid this by trying to be more careful. However, if we are aware of how many times we overcount, we can also use this to our advantage!

Example 15.2 How many diagonals are there in a regular polygon with $n \geq 3$ sides?

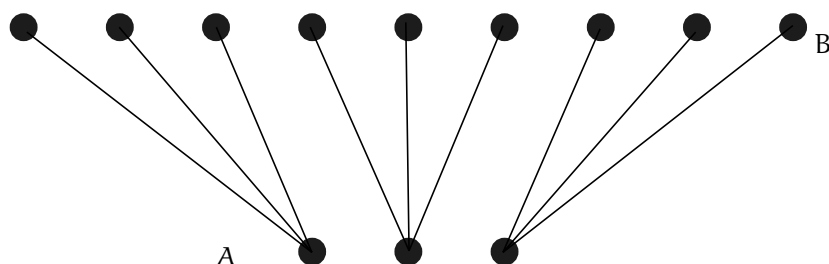
This was an exercise in the chapter on induction, but now we will give a different proof. A first attempt is as follows using the Multiplication Principle. Given two vertices v and w , we will let $\overline{vw}$ denote the line segment between them. Pick a corner v , there are n options. Then there are $n-3$ other corners w that you can pick to form a diagonal $\overline{vw}$ (since we have to exclude the vertex v and the two vertices next to it, since connecting to

those from v won't form a diagonal contained in the polygon). Thus, there are $n(n-3)$ ways of picking a corner v and then another corner w so that $\overline{vw}$ forms a diagonal.

However, this is *not* the final answer. Notice that in this algorithm, each diagonal actually gets counted twice, since for two nonadjacent corners v and w , $\overline{vw} = \overline{wv}$. What we have really counted is the number of ordered pairs (v, w) of corners that are not adjacent, and every diagonal corresponds to exactly two such pairs. Thus, if we divide $n(n-3)$ by two, we get the true number of diagonals, that is, $n(n-3)/2$. (And note that either n or $n-3$ is even!)

The general idea behind overcounting is as follows: suppose you want to count a set of things A . Suppose there is another set B that (a) is easier to count and (b) *to each* $a \in A$ *there correspond* m *elements of* B , that is, B and A are in m -to-one correspondence. This means we can pair each element of A with m elements of B so that every element of B belongs to exactly one pairing. If we call the set of these pairs P , then $m|A| = |P|$ (since every a is in exactly m pairings of the form $(a, b) \in P$), and $|P| = |B|$ since every $b \in B$ belongs in exactly one pairing. Thus, we get that $m|A| = |B|$.

In the previous example, we could match every diagonal d with two ordered pairs of nonadjacent corners; the total number of such ordered pairs is $n \cdot (n-3)$, and this is twice the number of diagonals. Thus, the number of diagonals is this number divided by 2. In the figure below, we show an example of how, to each element in a set A , there correspond 3 elements in a set B (and so $|B| = 3|A|$).



Example 15.3 How many ways are there to rearrange the letters in the words "orange" and "banana"?

By the Ordering Theorem, there are $6!$ ways of rearranging the letters in "orange" since there are six *distinct* letters, but for "banana" we have to be careful.

Suppose first that we treat each letter as an individual letter, that is, we look at the number of rearrangements of the word " $ba_1n_1a_2n_2a_3$ ". There are now 6 distinct characters, so there are a total of $6!$ ways of rearranging this word. This isn't the same thing as counting the number of ways of rearranging "banana", since if we change the a_1, a_2, a_3 in a rearrangement back into a 's and the n_1 and n_2 back into n 's, there are multiple ways of getting the same rearrangement of "banana." For example, "nanaba" can be obtained from " $n_1a_1n_2a_3ba_2$ " as well as " $n_2a_2n_1a_3ba_1$ ". However, to each arrangement of "banana", there are $3!$ ways of treating the a 's as distinct letters and $2!$ ways of treating the n 's as distinct letters so that they return the same rearrangement, and hence there are $3! \cdot 2!$ rearrangements of " $ba_1n_1a_2n_2a_3$ " that correspond to the same rearrangement of "banana" when we turn the n_i 's into n 's and the a_i 's into a 's. Thus, counting the number of rearrangements of " $ba_1n_1a_2n_2a_3$ " (which is $6!$) overcounts the number of rearrangements of "banana" by a factor of $3! \cdot 2!$. Hence, we just need to divide $6!$ by

$3! \cdot 2!$ to get the number of rearrangements of "banana", which is

$$\frac{6!}{3! \cdot 2!} = \frac{720}{12} = 60.$$

(Note that the original arrangement of the letters in "orange" counts as one of the $6!$ rearrangements.)

15.3 Counting subsets

We will now use the method of overcounting to count the number of subsets of a set S with $|S| = n$ which have a given size k . Because this number appears over and over again, we give it a special notation:

Definition 15.2 Let S be set with n distinct members, and let k be an integer such that $0 \leq k \leq n$. We let

$$\binom{n}{k}$$

denote the number of subsets of S which have exactly k members (and the above is read "n choose k" because it gives the number of ways we can choose k members from an n -element set). (If $k > n$, we set $\binom{n}{k} = 0$).

Theorem 15.5 For integers $0 \leq k \leq n$, we have

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}. \quad (15.3.1)$$

Liebeck gives a different proof in the book of this theorem, so feel free to look at that proof too. Recall that by convention we declare $0! = 1$.

Notice also that

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n(n-1)\dots(n-k+1)}{k(k-1)\dots 2 \cdot 1}. \quad (15.3.2)$$

For example $\binom{3}{2} = \frac{3!}{2!1!} = 3$, $\binom{6}{3} = \frac{6!}{3!3!} = \frac{6 \cdot 5 \cdot 4}{3 \cdot 2 \cdot 1} = 20$ and $\binom{n}{0} = 1$.

Proof. If $k = 0$, there is only one subset of size 0, which is the empty set, and in this case we see that $\frac{n!}{0!(n-0)!} = \frac{n!}{n!} = 1$. Now let $1 \leq k \leq n$ and S be a set of n elements. Consider the process where we pick an ordered list $(a_1, \dots, a_k)$ of k distinct elements from S , one at a time, to form a set $A = \{a_1, \dots, a_k\}$ of k elements. By Proposition 15.4, there are $\frac{n!}{(n-k)!}$ ways of picking the list $(a_1, \dots, a_k)$. However, this is not the same thing as counting the number of subsets of size k , since if we picked $a_1, \dots, a_k$ in a different order, they would still give rise to the same set $A = \{a_1, \dots, a_k\}$. What we have really counted is the number of ways of picking an *ordered* list of k elements, not a *set* of k elements. The number of orderings of $\{a_1, \dots, a_k\}$ is $k!$ by the Ordering Theorem. Thus, we have overcounted the number of subsets of size k by a factor of $k!$, hence if we divide $\frac{n!}{(n-k)!}$ by $k!$, we get that the number of subsets of size k is the right side of (15.3.1). ■

Note that it's perhaps not completely obvious that the right-hand side of (15.3.1) actually defines a whole number; however once we can identify it as the number of ways of doing something, it *has to be* a nonnegative integer!

This argument leads us to a new kind of proof, called a *combinatorial proof*, which is a way of proving a formula by showing that both sides of an equation count the same number of things, rather than trying to establish the equation directly via algebra. Let's start with an easy example:

Example 15.4 For $0 \leq k \leq n$ we have

$$\binom{n}{k} = \binom{n}{n-k}.$$

This can be easily shown using the formula (15.3.1), but we will prove it instead using one-to-one correspondence (that is, m -to-one correspondence with $m = 1$):

Note that if S is a set of size n and A is a subset of size k , then $S \setminus A$ is a set of size $n - k$. Conversely, if B is a set of size $n - k$, then $S \setminus B$ is a set of size k . Therefore to every set of size k we can associate a unique set of size $n - k$ and vice-versa. Thus the sets of size k and sets of size $n - k$ are in one-to-one correspondence, so the number of these sets is the same, that is, $\binom{n}{k} = \binom{n}{n-k}$. (If you need a bit more convincing, check that if $B = S \setminus A$, then $S \setminus B = A$.) Slightly more formally we describe the correspondence $A \leftrightarrow S \setminus A$ as a *bijection* between the sets of size k and the sets of size $n - k$.

Let's prove another identity where it may be less clear how to prove it algebraically.

Corollary 15.6 For $n \geq 0$, we have

$$2^n = \sum_{k=0}^n \binom{n}{k}.$$

Proof. Let S be a set of size n , let A be the set of subsets of S , and for $k = 0, \dots, n$, let A_k denote the subsets of S of size k , so A_k has size $\binom{n}{k}$ by Theorem 15.5. These sets are disjoint and their union is all of A , so they form a *partition* of A . Thus, the size of A is the sum of the sizes of the A_k . Hence, with $|A|$ denoting the size of A ,

$$|A| = |A_1| + \dots + |A_n| = \sum_{k=0}^n |A_k| = \sum_{k=0}^n \binom{n}{k}.$$

We also know by Proposition 15.2 that $|A| = 2^n$, and this proves the corollary. ■

We have surreptitiously introduced the notion of a partition above. More formally, a *partition* of a nonempty finite set A is a collection of subsets $A_1, \dots, A_m$ of A such that $\bigcup_{i=1}^m A_i = A$ and such that for $i \neq j$, $A_i \cap A_j = \emptyset$. In other words, every $x \in A$ belongs to *exactly one* A_i .

Suppose we are given n identical red objects laid out along a line, and we are told to *intersperse* m identical blue objects amongst them. (We are allowed to begin and end our new chain of objects with blue objects if we wish.) How many ways of doing this are there?

Proposition 15.7 – Linear interspersion principle. The number of ways of interspersing m identical blue objects amongst n identical red objects laid along a line is $\binom{n+m}{m}$.

Proof. Each linear arrangement combining the m blue objects and the n red objects corresponds to choosing a subset of size m (the blue objects) from a set of size $m + n$ (all the

objects). There are $\binom{n+m}{m}$ such choices, and hence $\binom{n+m}{m}$ ways of interspersing m identical blue objects amongst n identical red objects laid along a line. ■

Example 15.5 How many ways are there to express the number 9 as a sum of three nonnegative integers?

This uses a clever trick which reduces matters to the linear interspersion principle. (It might look intimidating to think it up on your own, but now you'll know it!).

Consider a sum $x + y + z = 9$ with $0 \leq x, y, z \leq 9$ and replace x, y, z with the number of 1's that add up to x, y and z respectively separated by + 's. For example, $4 + 3 + 2 = 9$ and $5 + 0 + 4$ correspond to

$$1\ 1\ 1\ 1 + 1\ 1\ 1 + 1\ 1 \quad \text{and} \quad 1\ 1\ 1\ 1\ 1 + + 1\ 1\ 1\ 1.$$

Thus, the number of solutions to $x + y + z = 9$ is exactly the number of ways to intersperse 2 + 's amongst 9 1's laid out in a row. This is the case $n = 9, m = 2$ of the linear interspersion principle, and the number of such solutions is thus $\binom{9+2}{2} = \binom{11}{2}$.

Example 15.6 How many ways are there to place 9 identical red balls in three boxes?

This is the previous example in disguise! Put the 9 balls in a line, and introduce two "dividers" to delineate the "gaps" between the three boxes. So, for example

$$B\ B\ B\ B\ | B\ | B\ B\ B\ B$$

denotes placing 4 balls in box 1, 1 ball in box 2 and 4 balls in box 3. The number of ways of inserting two dividers in a line of 9 balls is therefore $\binom{11}{2}$.

Now try your hand at combinatorial proofs in the following exercises. The second one will be a workshop problem.

■ **Exercise 15.1** Given $S = \{1, 2, \dots, n\}$ and $0 \leq j < n$, show that number of subsets of S of the form $\{a, a+1, \dots, a+j\} \subseteq S$ is $\binom{n}{2} + n$.

■ **Exercise 15.2** Let $1 \leq r \leq n$. Prove that

$$\binom{n+1}{r} = \binom{n}{r} + \binom{n}{r-1}. \quad (15.3.3)$$

(You may be familiar with this result if you know about *Pascal's triangle*.) First prove this by induction using (15.3.1). But then can you give another proof by counting a single quantity in two different ways?

Use this result to show that for $2 \leq r \leq n$,

$$\binom{n+2}{r} = \binom{n}{r} + 2\binom{n}{r-1} + \binom{n}{r-2}.$$

How can you further generalise these results?

15.4 Counting partitions

The binomial coefficient $\binom{n}{k}$ counts the number of ways to pick a subset of size k from a set of size n , or in other words, the number of ways of splitting a set of size n into a set of

size k and a set of size $n - k$. What if we wanted to count the number of ways to partition a set into more than two sets?

For example, let $A = \{1, 2, 3, 4, 5, 6\}$. Then $\{\{1, 2, 3\}, \{4, 5\}, \{6\}\}$ is a partition of A into sets of sizes 3, 2 and 1 respectively. Another one is $\{\{1, 4, 6\}, \{3, 5\}, \{2\}\}$. How many such partitions are there? One way to look at this is as follows. First we work out the number of ways to select 3 members of A for our first set: there are $\binom{6}{3}$ such ways. Each such choice leaves us with a set of 3 remaining members of A , from which we choose 2 for our second set, and there are $\binom{3}{2}$ ways to do this. Finally there is exactly one choice for the last set. So altogether there are

$$\binom{6}{3} \times \binom{3}{2} \times 1 = \frac{6!}{3!3!} \times \frac{3!}{2!1!} \times \frac{1!}{1!} = \frac{6!}{3!2!1!}$$

ways to form such a partition, by the Multiplication Principle.

Another example: again let $A = \{1, 2, 3, 4, 5, 6\}$ and suppose this time we want to partition A into non-interchangeable sets of sizes 2, 2 and 2. More concretely, suppose we are given a red, a blue and a green bin, and we want to calculate the number of ways of throwing 2 elements of A into the red bin, 2 elements of A into the blue bin and 2 elements of A into the green bin. (In the previous example there was no need to assign colours to the bins because they were always going to hold *different* numbers of members of A – the red bin was always going to hold 3 members of A , the blue bin 2 members and the green bin 1 member of A .) There are $\binom{6}{2}$ ways to throw 2 members of A into the red bin, $\binom{4}{2}$ ways of throwing 2 of the remaining 4 members of A into the blue bin, and $\binom{2}{2}$ ways of throwing the remaining 2 members of A into the green bin. So altogether there are

$$\binom{6}{2} \times \binom{4}{2} \times \binom{2}{2} = \frac{6!}{4!2!} \times \frac{4!}{2!2!} \times \frac{2!}{2!} = \frac{6!}{2!2!2!}$$

ways to form such a partition, by the Multiplication Principle.

What we are discussing is more formally described using the terminology of an *ordered partition*.

Definition 15.3 Let S be a nonempty finite set. An *ordered partition* of S is a sequence of sets $(A_1, \dots, A_k)$ that partitions S .

In the case of our second example, the two ordered partitions $(\{1, 2\}, \{3, 4\}, \{5, 6\})$ and $(\{3, 4\}, \{1, 2\}, \{5, 6\})$ are distinct because in the first case 1 and 2 are in the red bin while in the second case they are in the blue bin; we want to count these separately. In general we think of having k colours, and the set A_1 as being assigned colour 1, A_2 as being assigned colour 2, and so on.

Definition 15.4 Given $n \in \mathbb{N}$, $k \in \mathbb{N}$ with $k \geq 2$, and nonnegative integers $r_1, \dots, r_k$ such that $r_1 + \dots + r_k = n$, we denote the number of ordered partitions $(A_1, \dots, A_k)$ of a set S such that $|A_i| = r_i$ by

$$\binom{n}{r_1, \dots, r_k}.$$

Our examples show us that

$$\binom{6}{3, 2, 1} = \frac{6!}{3!2!1!}$$

and

$$\binom{6}{2, 2, 2} = \frac{6!}{2!2!2!}.$$

The following result should therefore not be too surprising.

Theorem 15.8 For $n \in \mathbb{N}$, $k \geq 2$ and nonnegative integers $r_1, \dots, r_k$ such that $r_1 + \dots + r_k = n$, we have

$$\binom{n}{r_1, \dots, r_k} = \frac{n!}{r_1! r_2! \dots r_k!}.$$

Again, we give a different proof from Liebeck.

Proof. We prove the above formula by induction on k . Our inductive hypothesis $P(k)$ for some integer $k \geq 2$, is that the formula in the statement of Theorem 15.8 holds for *all* $n \in \mathbb{N}$ and *all* $r_1, \dots, r_k$ such that $r_1 + \dots + r_k = n$.

Base case: If $k = 2$ and S is an arbitrary set of size n , $P(k)$ follows since the number of ordered partitions (A_1, A_2) of S , where $|A_1| = r_1$ and $|A_2| = r_2 = n - r_1$ is just the number of ways of choosing an r_1 -element subset of S which equals

$$\binom{n}{r_1} = \binom{n}{r_1, n - r_1}.$$

This proves the base case $P(2)$.

Inductive Step: Suppose the inductive hypothesis $P(k)$ holds for some $k \geq 2$. Again, we let S be an arbitrary set of size n . Picking an ordered partition $(A_1, \dots, A_{k+1})$ of S with $|A_i| = r_i$ for each integer $1 \leq i \leq k$ and $r_1 + \dots + r_{k+1} = n$ is the same as picking an r_{k+1} -member subset A_{k+1} of S , and, for each such subset, then picking an ordered partition $A_1, \dots, A_k$ of the remaining $(n - r_{k+1})$ -member set $S \setminus A_{k+1}$ with $|A_i| = r_i$ for integers $1 \leq i \leq k$. By the Multiplication Principle the number of ways of doing this is

$$\binom{n}{r_{k+1}} \times \binom{n - r_{k+1}}{r_1, \dots, r_k}$$

which, by the inductive hypothesis, is

$$\binom{n}{r_{k+1}} \times \frac{(n - r_{k+1})!}{r_1! \dots r_k!} = \frac{n!}{(n - r_{k+1})! r_{k+1}!} \frac{(n - r_{k+1})!}{r_1! \dots r_k!} = \frac{n!}{r_1! r_2! \dots r_{k+1}!},$$

showing that $P(k+1)$ holds and completing the inductive step.

Hence, as $P(2)$ holds and $P(k) \implies P(k+1)$ for each integer $k \geq 2$, we conclude that $P(k)$ holds for all such k by induction. ■

Take a good look at the above proof and check that the inductive hypothesis was indeed valid at the inductive step.

Example 15.7 Let's revisit the "banana" problem. Each arrangement of "banana" corresponds to an ordered partition (B, A, N) of the set $\{1, 2, \dots, 6\}$ with $|B| = 1, |A| = 3, |N| = 2$, where $i \in B$ if the i th letter in the rearrangement is a "b", $i \in A$ if the i th letter is an "a", and $i \in N$ if the i th letter is an "n". Thus, the total number of rearrangements is again

$$\binom{6}{3, 2, 1} = \frac{6!}{3! 2! 1!} = 60.$$

Example 15.8 How many ways are there to choose from a set of size 10 three disjoint subsets of sizes 5, 3, and 2 respectively? What about sizes 4, 4, and 2 respectively?

Let $S = \{1, 2, \dots, 10\}$. There are $\binom{10}{5, 3, 2}$ ways of choosing an ordered partition of S into

a first set of size 5, a second set of size 3, and a third set of size 2 by Theorem 15.8.

For the second part, there are $\binom{10}{4,4,2}$ ways of picking an *ordered* partition with sets of sizes 4, 4, and 2. But what if we instead wanted the number N of *unordered* partitions $\{A_1, A_2, A_3\}$ where $|A_1| = |A_2| = 4$ and $|A_3| = 2$? Given such a partition, there are $2!$ ordered partitions we can make from it, (by swapping the two sets A_1 and A_2 of the same size 4), so that the first two sets have size 4 and the last has size 2. Thus, counting the number of ordered partitions overcounts the number of unordered partitions by a factor of 2, so the number of partitions is $N = \binom{10}{4,4,2}/2$.

15.5 The Binomial and Multinomial Theorems

A very natural place that binomial coefficients appear (and the reason why they are called binomial coefficients) is in the following theorem, with which you are probably already familiar with from school:

Theorem 15.9 — The Binomial Theorem. Let $n \in \mathbb{N}$ and $a, b \in \mathbb{C}$. Then^a

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}. \quad (15.5.1)$$

^aLiebeck assumes $a, b \in \mathbb{R}$, but this is not necessary.

As an application, let's give a second (faster) proof of Corollary 15.6: by the Binomial Theorem,

$$2^n = (1 + 1)^n = \sum_{k=0}^n \binom{n}{k} 1^k 1^{n-k} = \sum_{k=0}^n \binom{n}{k}.$$

Proof. We have

$$(a + b)^n = \underbrace{(a + b)}_1 \cdot \underbrace{(a + b)}_2 \cdots \underbrace{(a + b)}_n.$$

When we multiply out this product, we will get a sum of terms of the form $a^k b^{n-k}$ for $0 \leq k \leq n$. How many of each of these do we get? Each bracket contributes either an a or a b in the final answer, and the number of $a^k b^{n-k}$ terms in this final answer will be precisely the number of ways of picking k brackets from the brackets labelled $1, 2, \dots, n$ above. That is, it is $\binom{n}{k}$. In other words, the coefficient of $a^k b^{n-k}$ in the expansion is exactly $\binom{n}{k}$. ■

The binomial theorem is a special case of the following theorem which is proved in much the same way:

Theorem 15.10 — The Multinomial Theorem. Let $n \in \mathbb{N}$ and $x_1, \dots, x_k \in \mathbb{C}$. Then

$$(x_1 + \cdots + x_k)^n = \sum \binom{n}{r_1, \dots, r_k} x_1^{r_1} \cdots x_k^{r_k} \quad (15.5.2)$$

where the sum is taken over all $r_i \in \{0, 1, \dots, n\}$ such that $r_1 + \cdots + r_k = n$.

Proof. Let's number the parentheses in the product:

$$(x_1 + \cdots + x_k)^n = \underbrace{(x_1 + \cdots + x_k)}_1 \cdot \underbrace{(x_1 + \cdots + x_k)}_2 \cdots \underbrace{(x_1 + \cdots + x_k)}_n.$$

When we multiply out this product, we will get terms of the form $x_1^{r_1} \cdots x_k^{r_k}$ where $r_1 + \cdots + r_k = n$. How many of these terms do we get? We can form such a product by picking a partition $A_1, \dots, A_k$ of $\{1, 2, \dots, n\}$ with $|A_i| = r_i$ and then picking x_j from A_i to include in our $x_1^{r_1} \cdots x_k^{r_k}$ term if $j \in A_i$. Thus, the number of times $x_1^{r_1} \cdots x_k^{r_k}$ appears when we multiply out the above product is the same as counting the number of ordered partitions $(A_1, \dots, A_k)$ of $\{1, 2, \dots, n\}$ with $|A_i| = r_i$, which is $\binom{n}{r_1, \dots, r_k}$. ■

Example 15.9 What is the coefficient of x in $(1 + x + \frac{1}{x})^5$?

Let $a = 1$, $b = x$ and $c = \frac{1}{x}$. When we expand $(a + b + c)^5$, the terms of the form $a^{r_1} b^{r_2} c^{r_3}$ (where $r_i \geq 0$ and $r_1 + r_2 + r_3 = 5$) that equal x are of the form $a^4 b$, $a^2 b^2 c$, and $b^3 c^2$, so we just need to add together the coefficients of these terms, which by the multinomial theorem are

$$\binom{5}{4,1,0} + \binom{5}{2,2,1} + \binom{5}{0,3,2} = \frac{5!}{4!0!} + \frac{5!}{2!2!1!} + \frac{5!}{0!3!2!} = 5 + \frac{120}{4} + \frac{120}{12} = 5 + 30 + 10 = 45.$$

15.6 Inclusion-Exclusion

Suppose we want to count how many numbers between 1 and 100 are divisible by either 3 or 5. A first attempt might be to say, well, there are $99/3 = 33$ numbers divisible by 3 and $100/5 = 20$ numbers that are divisible by 5, so the numbers that are divisible by 3 or 5 is the sum of these two numbers, i.e. $33 + 20 = 53$. However, we again run into an *overcounting* problem, since some numbers are divisible both by 3 and 5 and thus are being counted twice. We can't just divide our answer by 2 as before, since there are plenty of numbers that are divisible by 3 but not 5 and vice versa. The way to overcome this is by using the *inclusion-exclusion principle*:

Theorem 15.11 — Inclusion-Exclusion Principle. Let $n \geq 2$ and suppose $A_1, A_2, \dots, A_n$ are finite sets. Let

$$N_1 = |A_1| + |A_2| + \cdots + |A_n|,$$

$$N_2 = |A_1 \cap A_2| + |A_1 \cap A_3| + |A_2 \cap A_3| + \cdots + |A_{n-1} \cap A_n|,$$

and similarly, let

$$N_k = \sum_{1 \leq i_1 < i_2 < \cdots < i_k \leq n} |A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}|$$

be the sum of the sizes of the intersections of all collections of k different A_i 's, so that

$$N_n = \left| \bigcap_{k=1}^n A_k \right|.$$

Then

$$|A_1 \cup \cdots \cup A_n| = N_1 - N_2 + N_3 - N_4 + \cdots + (-1)^{n+1} N_n = \sum_{k=1}^n (-1)^{k+1} N_k. \quad (15.6.1)$$

In particular, for finite sets A, B, C ,

$$|A \cup B| = |A| + |B| - |A \cap B|,$$

and

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

Proof. Let $A = A_1 \cup \dots \cup A_n$, and list the elements as $A = \{x_1, \dots, x_M\}$ where $M = |A|$. Notice that when we compute $N_1 = |A_1| + \dots + |A_n|$, this is the same as computing for each x_k the number of sets A_i that x_k appears in, and then summing over k . Similarly, N_2 is the same as adding together over all x_k the number of times x_k appears in some pair $A_i \cap A_j$, and so on. Thus, the sum on the right of (15.6.1) is the same as computing the sum over each x_k , how many times x_k appears in one set A_i minus how many times it appears in the intersection of two sets, and so on, and then adding over all $x_k \in A_1 \cup \dots \cup A_n$. What we will do is show that for each k , this number is 1.

Suppose x_k appears in each of the sets $A_{k_1}, \dots, A_{k_m}$ for some $1 \leq k_1 < k_2 < \dots < k_m \leq n$, and none of the others. This means x_k appears in exactly m sets, so its contribution to N_1 is exactly m . It also appears in $\binom{m}{2}$ many pairs $A_{k_i} \cap A_{k_j}$ with $1 \leq i < j \leq m$, so it contributes $\binom{m}{2}$ to N_2 . Similarly, x contributes $\binom{m}{k}$ to N_k for each $k \leq n$ (and recall that $\binom{m}{k} = 0$ for $k > m$). Thus, by the Binomial Theorem, the total contribution of x to the right side of (15.6.1) is

$$\sum_{k=1}^m \binom{m}{k} (-1)^{k+1} = 1 + \sum_{k=0}^m \binom{m}{k} (-1)^{k+1} = 1 - \sum_{k=0}^m \binom{m}{k} (-1)^k = 1 - (1-1)^m = 1.$$

■

This proof is quite abstract, but using it is a lot easier, which we demonstrate in the following example.

Example 15.10 Let's return to the example of counting how many natural numbers at most 100 are divisible by 5 or 3. Let A be those numbers divisible by 3 and B those numbers which are divisible by 5. Then as we showed before, $|A| = 33$ and $|B| = 20$. Note that the set we want to count is $A \cup B$, which are those numbers either in A or in B (that is, divisible by 3 or divisible by 5). Also note that $A \cap B$ are those numbers divisible by both 3 and 5, that is, divisible by 15, of which there are only 6. Thus,

$$|A \cup B| = |A| + |B| - |A \cap B| = 33 + 20 - 6 = 47.$$

15.7 Exercises

- **Exercise 15.3** How many ways are there for n people to stand in a circle?
- **Exercise 15.4** How many ways are there to partition a set of size n into two subsets (one of which can be empty)?
- **Exercise 15.5** Determine how many ways there are to distribute 9 sweeties in 4 boxes if
 - (a) The sweeties and boxes are distinct.
 - (b) The sweeties are identical and the boxes are distinct.
- **Exercise 15.6** How many numbers at most 200 are divisible by

- (a) 5 or 6?
- (b) 3,4 or 5?
- (c) 3,5, or 6?

■ **Exercise 15.7** Use a combinatorial argument to prove that for $2 \leq k \leq n$,

$$\binom{k}{2} + \binom{n-k}{2} + k(n-k) = \binom{n}{2}.$$

■ **Exercise 15.8** Use a combinatorial argument to show

$$\binom{nk}{2} = k \binom{n}{2} + n^2 \binom{k}{2}.$$

■ **Exercise 15.9** Give two proofs that $\binom{2n}{n}$ is even.

■ **Exercise 15.10** If $n \geq 5$ distinct objects are arranged in a circle, how many ways are there to choose three of these n objects so that no two of them are next to each other?

■ **Exercise 15.11** Give both an algebraic argument and a combinatorial argument to show (for $0 \leq k \leq n$)

$$k \binom{n}{k} = n \binom{n-1}{k-1}$$

in two different ways. (**Hint for the combinatorial proof:** For a set S with $|S| = n$, the left side is the same as counting the number of pairs (a, A) where $a \in A \subseteq S$ and $|A| = k$. Can you show the right side counts the same set in a different way?)

■ **Exercise 15.12** Use a combinatorial argument to show

$$\sum_{k=0}^n k \binom{n}{k} = n2^{n-1}.$$

■ **Exercise 15.13 (Challenging!)** How many subsets does the set $\{1, 2, \dots, n\}$ have that contain no two consecutive integers? **Hint:** maybe compute this for the first few values of n to see if you can spot a pattern.

